

GUNTURU DILIP KUMAR

Senior Embedded Firmware Engineer | SIL-4 Safety Systems | 9+ Years

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PROFESSIONAL SUMMARY

Results-driven Senior Embedded Firmware Engineer with 9+ years of hands-on experience designing and certifying safety-critical embedded systems for the Indian Railways and automotive sectors. Spearheaded firmware architecture and SIL-4 certification activities for India's KAVACH Automatic Train Protection programme — one of the most rigorous railway safety deployments globally. Deep expertise in MISRA-C:2023, LDRA static analysis, IBM ELM requirements traceability, STM32 / PIC32MZ platforms, and multi-protocol communication stacks (CAN, UART, TCP/IP). Recognised twice for exceptional delivery under safety-certification timelines.

TECHNICAL EXPERTISE

Languages	Embedded C (Expert) · C# (Familiar)
Microcontrollers	STM32 (F/H/L series) · PIC32MZ/MX · LPC2148 · ATmega328P · ESP8266 · 8051
Protocols	CAN · SPI · I2C · UART · RS-232 / RS-422 / RS-485 · TCP/IP · UDP · RF
Safety & Quality	SIL-4 · MISRA-C:2012/2023 · LDRA Testbed (Static)
IDEs & Tools	STM32CubeIDE · IAR Embedded Workbench · MPLAB IDE · Visual Studio · Git
Lifecycle / ALM	IBM ELM (DOORS Next, ETM) · 4-way Requirements Traceability · Change Management
Architecture	Dual-Channel Safety Design · Real-Time Deterministic Execution · Software & System Architecture

PROFESSIONAL EXPERIENCE

Senior Embedded Firmware Engineer

Jul 2020 – Present

Lotus Wireless Technologies Pvt. Ltd · Visakhapatnam, AP

► KAVACH — SIL-4 Automatic Train Protection System (Anti Collision Project)

- Architected end-to-end System & Software Architecture for real-time STM32-based KAVACH safety modules, establishing deterministic execution flows compliant with EN 50128 / IEC 61508 SIL-4 requirements.
- Achieved 100% MC/DC coverage across all safety-critical functions using LDRA TBrn dynamic analysis — a mandatory certification gate for SIL-4 — delivering zero coverage deficiencies on final audit.
- Enforced MISRA-C:2023 across the entire codebase; formally documented and safety-manager-reviewed every approved deviation via IBM ELM, reducing violation count measurably release-over-release.
- Implemented four-way requirements traceability chain (System Req → Software Req → Source Code → Test Cases) in IBM ELM, enabling zero-deficiency safety case audits.
- Led firmware team coordination across hardware, system architecture, and V&V groups; introduced structured code review gates that cut post-review defect escapes significantly.
- Managed Git branching strategy and release tagging to support concurrent SIL-4 development and Verification & Validation activities without integration conflicts.

► WAG-7 / WAG-9 Locomotive Control (DPWCS)

- Developed RF-based device control and monitoring firmware for WAG-7 and WAG-9 electric locomotives operating in high-EMI traction environments with strict uptime requirements.
- Implemented multi-protocol communication stack (CAN, I2C, RS-232, RS-422, TCP/IP) on PIC32MZ2048EFH144, processing real-time locomotive telemetry reliably under load.

- ▶ Built hardware-datasheet-verified communication drivers with built-in error counters and diagnostic logging, directly reducing field maintenance response times.

▶ **ESMON — Energy & Speed Monitoring System**

- ▶ Designed and developed a pulse-generator-based embedded device for precise speed and distance measurement, delivering high accuracy in field conditions.
- ▶ Implemented robust diagnostic logging and error-counter framework, enabling proactive maintenance and rapid fault isolation by field teams.

Embedded Engineer

Sep 2019 – Jul 2020

Laghima Technologies · Visakhapatnam, AP

- ▶ Optimised AIS-140 compliant GPS tracking firmware on LPC2148 + SIM808, improving field-device stability across a deployed fleet of tracking units.
- ▶ Designed OBD-based automotive communication interfaces integrating vehicle diagnostic protocols for real-time fleet telemetry and health monitoring.
- ▶ Developed GSM/GPRS-connected fleet management firmware with NMEA GPS parsing, enabling live vehicle tracking and event-based reporting.
- ▶ Diagnosed and resolved intermittent field failures through systematic root-cause analysis, improving device uptime in live deployments.

Junior Research Engineer

Oct 2016 – Jul 2019

Academy of Robotics Pvt. Ltd · Visakhapatnam, AP

- ▶ Designed and deployed IoT-based embedded systems (ESP8266, ATmega328P, Raspberry Pi) for research automation — covering schematic design, PCB bring-up, firmware, and field deployment.
- ▶ Delivered end-to-end embedded solutions for multiple client projects; authored detailed technical documentation and milestone reports reviewed by stakeholders.

Embedded & Technical Trainee

Nov 2015 – Jul 2016

Embedded Wings · Visakhapatnam, AP

- ▶ Developed embedded applications on PIC16 and 8051 microcontrollers; supported circuit debugging, firmware testing, and technical documentation.

EDUCATION

B.Tech — Electronics & Communication Engineering

2015

Avanthi's Research and Technological Academy · Aggregate: 70.45%